BSc (Hons) in Information Technology

AF-PROJECT – SE3040

Sri Lanka Institute of Information Technology

Application Framework (SE3040)

2026

Ecocycle

Group SE-83

Deployment Report

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1. Introduction :

This deployment report describes the cloud deployment process of the E-Waste Platform MERN stack application. The backend RESTful API was deployed using Render, while the React frontend application was deployed using Vercel. The purpose of this deployment is to host the application in a production-ready cloud environment and make it accessible publicly via live URLs.

2. Backend Deployment Platform and Setup Steps:

Platform Used:

- **Render** – Cloud Platform for Node.js backend hosting

Deployment Steps:

1. Pushed backend source code to GitHub repository.
2. Logged into Render dashboard.
3. Created a new **Web Service**.
4. Connected GitHub repository to Render.
5. Selected backend repository branch (main).

6. Configured Render deployment settings:
    - Root Directory: Backend
    - Build Command: npm install
    - Start Command: npm start
7. Added required environment variables in Render.
8. Deployed backend service successfully.
9. Verified backend deployment through Render live URL.

3. Frontend Deployment Platform and Setup Steps :

Platform Used

- **Vercel** – Cloud Platform for React frontend hosting

Deployment Steps:

1. Pushed frontend source code to GitHub repository.
2. Logged into Vercel dashboard.
3. Imported GitHub repository.
4. Configured frontend project settings:
    1. Framework Preset: Vite
    2. Root Directory: Frontend
5. Added frontend environment variable:
    1. VITE_API_BASE_URL
6. Deployed frontend application.
7. Verified frontend deployment through Vercel live URL.

4. Live Deployment URLs :

Frontend Application : https://project-eco-cycle.vercel.app

Backend API : https://project-ecocycle.onrender.com

5. Environment Variables Used :

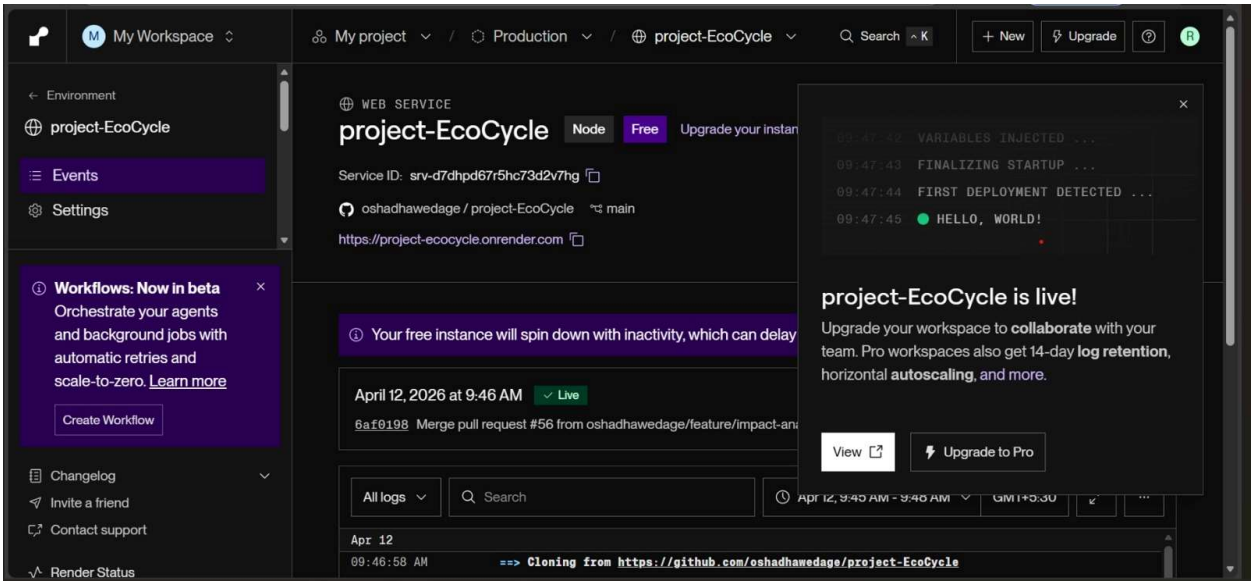
Backend Environment Variables :

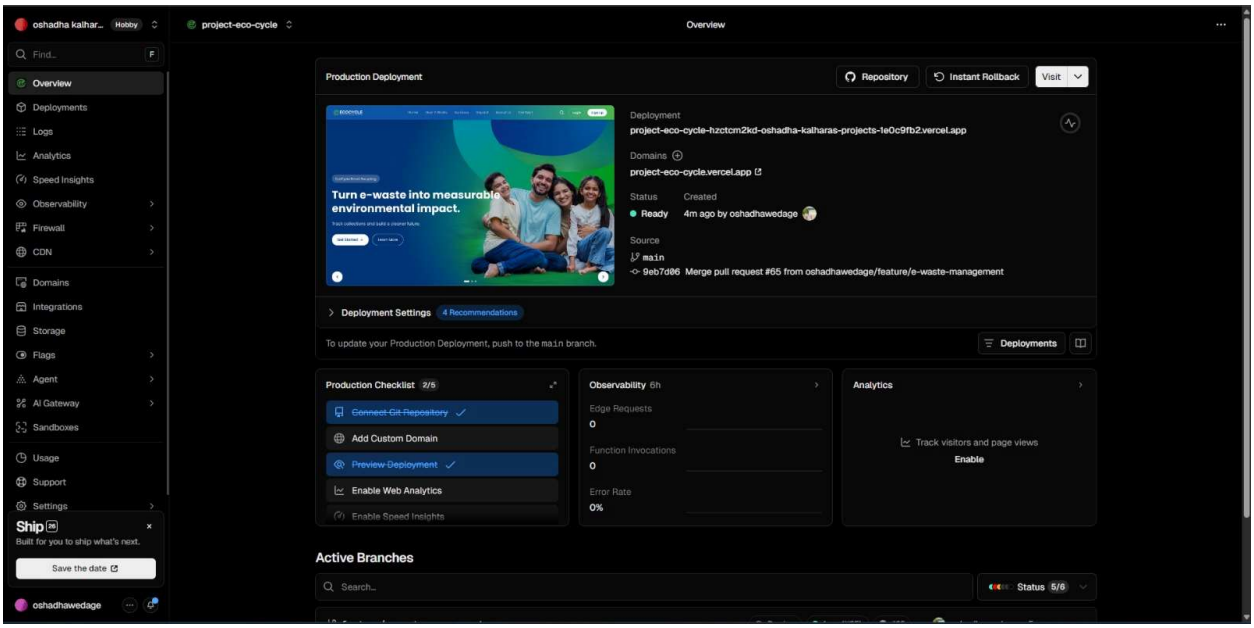
Variable Name	Purpose
MONGO_URI	MongoDB Atlas connection string used to connect the backend to the cloud database.
JWT_SECRET	Secret key used for generating and validating JWT authentication tokens.
JWT_EXPIRES	Defines JWT token expiration duration.
APP_NAME	Stores application name for internal use and email templates.
CLIENT_URL	Stores frontend application URL allowed for CORS requests.
EMAIL_HOST	SMTP server host for email service.
EMAIL_PORT	SMTP server port number.
EMAIL_USER	Email account username for SMTP authentication.
EMAIL_PASS	Email account password/app password for SMTP authentication.
EMAIL_FROM	Sender email configuration for outgoing emails.
OTP_EXPIRES_MIN	Defines OTP expiration duration in minutes.
ADMIN_REGISTER_KEY	Secret key used for secure admin registration.

Frontend Environment Variables:

Variable Name	Purpose
VITE_API_BASE_URL	Stores deployed backend API URL used by frontend for API requests.

6. Deployment Evidence :

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Hobby
project-eco-cycle
Overview
Find...
Overview
Deployments
Logs
Analytics
Speed Insights
Observability
Firewall
CDN
Domains
Integrations
Storage
Flags
Agent
AI Gateway
Sandboxes
Usage
Support
Settings
Ship
Built for you to ship what's next.
Save the date
oshadhawedage
Production Deployment
Repository
Instant Rollback
Visit
Turn e-waste into measurable environmental impact.
Deployment
project-eco-cycle-hzctcm2kd-oshadha-kalharas-projects-1e0c9fb2.vercel.app
Domains
project-eco-cycle.vercel.app
Status
Created
Ready
4m ago by oshadhawedage
Source
main
9eb7d06 Merge pull request #65 from oshadhawedage/feature/e-waste-management
Deployment Settings
4 Recommendations
To update your Production Deployment, push to the main branch.
Deployments
Production Checklist 2/5
Connect Git Repository
Add Custom Domain
Preview Deployment
Enable Web Analytics
Enable Speed Insights
Observability 6h
Edge Requests
0
Function Invocations
0
Error Rate
0%
Analytics
Track visitors and page views
Enable
Active Branches
Search...
Status 5/6